

Software Requirements Specification

Online Examination Portal

Document Control

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the Phase-1 requirements for the Online Examination Portal. It provides a common reference for requirements, validation, architecture, implementation, and review.

1.2 Scope

The portal supports students, teachers, and administrators in conducting and managing objective-type online examinations. Phase 1 covers authentication, question-bank management, examination configuration, examination attempts, automatic evaluation, result release, reporting, administration, security, and validation planning.

1.3 Intended Audience

- Students
- Teachers
- Administrators
- Developers and designers
- Testers and reviewers

1.4 Definitions

Term	Meaning
FR	Functional Requirement — a behavior or service the system shall provide.
NFR	Non-Functional Requirement — a measurable quality, constraint, or system property.
RTM	Requirements Traceability Matrix.
UC	Use Case.
VAL	Validation identifier used by the Test Plan.

1.5 Requirements Quality Method

Requirements are written using a consistent 'The system shall...' form. Each requirement is intended to be clear, unambiguous, concise, consistent, verifiable/measurable where applicable, feasible, and traceable to a use case or validation activity. This follows the requirements-engineering emphasis of ISO/IEC/IEEE 29148:2018.

1.6 Problem Statement Analysis

- Traditional examination activities can require separate processes for question preparation, examination scheduling, answer collection, evaluation, and result publication.
- The portal brings these activities into one controlled workflow with role-based access.
- The system must maintain consistency of the examination question set while allowing question order to be shuffled.
- The requirements must also make timing, answer handling, evaluation, security, and result release precise enough to test.

1.7 Feasibility Analysis

1.7.1 Technical Feasibility

- The required functions use standard web-application concepts such as authentication, database-backed question management, examination workflows, automatic evaluation, and reporting.
- The Phase-1 design can be implemented incrementally using modular services and a database-backed application.

1.7.2 Operational Feasibility

- The workflow separates Student, Teacher, and Administrator responsibilities.
- The defined use cases provide a clear basis for user interaction and validation.

1.7.3 Resource and Schedule Feasibility

- The project is suitable for an academic Phase-1 implementation using standard development and testing resources.
- Requirements, validation, and design can be completed before implementation and integration work.

1.7.4 Security Feasibility

- Authentication, authorization, input validation, session controls, protected examination data, and audit logging can be validated through planned security tests.
- Penetration testing is planned for the testing phase; it is not claimed as completed in Phase 1.

2. Overall Description

2.1 Product Perspective

The portal is a role-based online examination system. The logical design uses a presentation layer, application services, data persistence, and supporting authentication/evaluation/result services.

2.2 Product Functions

- The system shall allow students to log in using valid credentials.
- The system shall allow teachers to log in using valid credentials.
- The system shall allow the administrator to manage users and assign or manage system roles according to authorized permissions.
- The system shall allow teachers to create examinations using the configured examination settings.
- The system shall allow teachers to upload questions to the question bank.
- The system shall allow teachers to assign a difficulty level to each question.
- The system shall allow teachers to edit or remove questions before the associated examination is published.
- The system shall allow teachers to configure the examination duration and availability period.

- The system shall randomly select the configured number of questions from the predefined question pool when an examination is initialized.
- The system shall provide the same selected set of questions to all students attempting the same examination.
- The system shall present the selected examination questions in shuffled order to students.
- The system shall display the questions associated with the student's active examination attempt.
- The system shall allow students to select and save answers during an examination attempt.
- The system shall automatically submit an examination attempt when the configured examination duration expires.
- The system shall allow students to manually submit an examination before the configured examination duration expires.
- The system shall automatically evaluate objective questions using their configured correct answers.
- The system shall calculate the examination score from the evaluated answers.
- The system shall securely store examination attempts and their submitted answers.
- The system shall automatically release examination results at the configured result-release time.
- The system shall allow authorized teachers and administrators to view examination reports.
- The system shall display examinations that are currently available to the logged-in student according to the configured availability period.

2.3 Confirmed Examination Rules

- Objective-type questions only.
- Questions are randomly selected from the predefined question pool.
- All students attempting the same examination receive the same selected question set.
- Selected questions are presented in shuffled order.
- Students can revisit and change answers before submission.
- Teachers can edit or remove questions only before the examination is published.
- No notification feature is included.
- Teachers and authorized administrators can view examination reports.
- Results are automatically released at the configured result-release time.
- The teacher configures the examination duration and availability period.

3. Specific Requirements

3.1 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow students to log in using valid credentials.
FR-02	The system shall allow teachers to log in using valid credentials.
FR-03	The system shall allow the administrator to manage users and assign or manage system roles according to authorized permissions.
FR-04	The system shall allow teachers to create examinations using the configured examination settings.
FR-05	The system shall allow teachers to upload questions to the question bank.
FR-06	The system shall allow teachers to assign a difficulty level to each question.
FR-07	The system shall allow teachers to edit or remove questions before the associated examination is published.
FR-08	The system shall allow teachers to configure the

	examination duration and availability period.
FR-09	The system shall randomly select the configured number of questions from the predefined question pool when an examination is initialized.
FR-10	The system shall provide the same selected set of questions to all students attempting the same examination.
FR-11	The system shall present the selected examination questions in shuffled order to students.
FR-12	The system shall display the questions associated with the student's active examination attempt.
FR-13	The system shall allow students to select and save answers during an examination attempt.
FR-14	The system shall automatically submit an examination attempt when the configured examination duration expires.
FR-15	The system shall allow students to manually submit an examination before the configured examination duration expires.
FR-16	The system shall automatically evaluate objective questions using their configured correct answers.
FR-17	The system shall calculate the examination score from the evaluated answers.
FR-18	The system shall securely store examination attempts and their submitted answers.
FR-19	The system shall automatically release examination results at the configured result-release time.
FR-20	The system shall allow authorized teachers and administrators to view examination reports.
FR-21	The system shall display examinations that are currently available to the logged-in student according to the configured availability period.
FR-22	The system shall allow administrators to log in using valid credentials before accessing administrative functions.
FR-23	The system shall allow authorized administrators to manage approved system-level settings.
FR-24	The system shall allow authorized administrators to manage system access permissions according to defined roles.

3.2 Non-Functional Requirements

NFR-03 and NFR-06 are the measurable targets carried in the Phase-1 handoff. The remaining NFRs below are proposed team requirements added to make the quality requirements complete and testable; they should be approved as part of final requirements review.

ID	Category	Requirement	Status
NFR-01	Security	The system shall require authenticated access for protected functions and enforce role-based authorization for Student, Teacher, and Administrator actions.	Proposed team NFR
NFR-02	Reliability	The system shall preserve submitted examination answers and attempt data without loss during normal operation.	Proposed team NFR
NFR-03	Performance	The system shall respond to common user actions within 3 seconds under normal load.	Proposed measurable target in handoff
NFR-04	Availability	The system shall remain	Proposed team NFR

		available to authorized users throughout a configured examination availability period, except during planned maintenance.	
NFR-05	Usability	The system shall provide clear navigation, readable instructions, and understandable feedback for major user actions.	Proposed team NFR
NFR-06	Learnability	A first-time student shall be able to start and submit an examination within 5 minutes without assistance.	Proposed measurable target; approval required
NFR-07	Maintainability	The system shall use modular components and documented interfaces so individual modules can be modified or tested with minimal impact on unrelated modules.	Proposed team NFR
NFR-08	Compatibility	The web portal shall support the current major version of commonly used desktop web browsers used by target users.	Proposed team NFR
NFR-09	Testability	Each functional and non-functional requirement shall have an associated validation or test method with observable acceptance criteria.	Proposed team NFR
NFR-10	Scalability	The system shall support an increase in concurrent examination users without changing the defined examination workflow.	Proposed team NFR
NFR-11	Data Integrity	The system shall maintain the integrity and consistency of questions, answers, attempts, scores, and released results throughout their lifecycle.	Proposed team NFR
NFR-12	Auditability	The system shall record sufficient information about important administrative and examination actions to support review and troubleshooting.	Proposed team NFR

3.3 External Interface Requirements

- User interface: browser-based screens for login, question management, examination attempt, submission, results, reports, and administration.
- Authentication interface: validates credentials before protected operations.
- Data interface: persistent storage for users, questions, examinations, attempts, answers, scores, and results.
- Service interface: application services expose controlled operations for examination, evaluation, and result workflows.

4. Security Requirements

4.1 Security Objectives

- Protect examination data, answers, attempts, and results from unauthorized access or modification.
- Ensure that users can perform only the operations allowed by their assigned role.
- Prevent premature result disclosure and unauthorized access to active examination data.
- Provide sufficient audit information for security-relevant administrative actions.

4.2 Security Requirements

ID	Security Requirement	Related NFR
SEC-01	The system shall authenticate users before granting access to protected functions.	NFR-01
SEC-02	The system shall enforce role-based authorization at protected operations for Student, Teacher, and Administrator roles.	NFR-01
SEC-03	The system shall prevent unauthorized users from accessing another user's examination attempt or result.	NFR-01
SEC-04	The system shall validate user-supplied input before processing or storing it.	NFR-01
SEC-05	The system shall protect submitted examination attempts and answers from unauthorized modification.	NFR-11
SEC-06	The system shall prevent results from being displayed before the configured result-release time.	NFR-01
SEC-07	The system shall expire inactive authenticated sessions according to the configured session policy.	NFR-01
SEC-08	The system shall record security-relevant administrative actions for audit and review.	NFR-12
SEC-09	The system shall prevent unauthorized users from accessing examination questions before the associated examination is published.	NFR-01
SEC-10	The system shall prevent error responses from exposing credentials, sensitive data, stack traces, or other internal implementation details.	NFR-01

5. UML Use Case Model

The use-case model identifies the main human actors and supporting system services. The Notification Service use case is excluded because the finalized Phase-1 rules explicitly state that no notification feature is included.

Figure 1. Online Examination Portal — Updated Use Case Diagram (includes UC-A06 Administrator Login)

6. Requirements Traceability Matrix

Requirement	Use Case / Quality Area	Validation
FR-01	UC-S01	VAL-F01
FR-02	UC-T01	VAL-F01
FR-03	UC-A01, UC-A02, UC-A03, UC-A05	VAL-F01, VAL-F15, VAL-F16, VAL-F18
FR-04	UC-T05	VAL-F04
FR-05	UC-T02	VAL-F02
FR-06	UC-T03	VAL-F03
FR-07	UC-T04	VAL-F13
FR-08	UC-T06	VAL-F04
FR-09	UC-S03, UC-X02	VAL-F05
FR-10	UC-S03, UC-X02	VAL-F05
FR-11	UC-S03, UC-S04	VAL-F06
FR-12	UC-S04	VAL-F06
FR-13	UC-S05	VAL-F12
FR-14	UC-S06	VAL-F07
FR-15	UC-S06	VAL-F07
FR-16	UC-X03	VAL-F08
FR-17	UC-X03	VAL-F08
FR-18	UC-S06	VAL-S11
FR-19	UC-S07, UC-X04	VAL-F09, VAL-F10
FR-20	UC-T07, UC-A04	VAL-F14
FR-21	UC-S02	VAL-F11
NFR-01	Quality attribute / cross-cutting	VAL-N01
NFR-02	Quality attribute / cross-cutting	VAL-N02
NFR-03	Quality attribute / cross-cutting	VAL-N03
NFR-04	Quality attribute / cross-cutting	VAL-N04
NFR-05	Quality attribute / cross-cutting	VAL-N05
NFR-06	Quality attribute / cross-cutting	VAL-N06
NFR-07	Quality attribute / cross-cutting	VAL-N07
NFR-08	Quality attribute / cross-cutting	VAL-N08
NFR-09	Quality attribute / cross-cutting	VAL-N09
NFR-10	Quality attribute / cross-cutting	VAL-N10
NFR-11	Quality attribute / cross-cutting	VAL-N11
NFR-12	Quality attribute / cross-cutting	VAL-N12
FR-22	UC-A06	VAL-F20
FR-23	UC-A03	VAL-F21
FR-24	UC-A05	VAL-F22

7. Requirement Status and Notes

- FR-22 explicitly covers Administrator Login and is mapped to UC-A06 and VAL-F20.
- FR-01 to FR-24 are the finalized functional requirements from the Phase-1 requirements handoff.
- FR-21 is the finalized requirement for UC-S02 — View Available Examinations.
- No notification feature is included; therefore a notification use case is not treated as a final functional requirement.
- NFR-03 and NFR-06 retain the measurable targets from the Phase-1 handoff; NFR-06 requires team approval before being treated as final.
- The additional NFRs are proposed quality requirements intended to make the SRS and Test Plan complete and measurable.
- FR-23 and FR-24 explicitly cover administrator system settings and access management, so UC-A03 and UC-A05 are not orphaned.
- FR-20 covers authorized teachers and administrators, matching UC-T07 and UC-A04.
- SEC-09 covers examination-question confidentiality before publication; SEC-10 covers safe error handling.